

GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY

Faculty of Computing

FINAL REPORT

ESSENTIALS OF ARTIFICIAL INTELLEGENCE

Project Details		
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Project Name	SafeMail AI – AI Based Phishing Email Detector
Submission Date	09/10/2026

Table of Contents

1. Introduction 3

2. Problem Background 4

3. Problem Statement 5

4. Objectives 5

4.1 Main Objective 5

4.2 Specific Objectives 5

5. Selected AI Techniques 6

5.1 Rule-Based Reasoning 6

5.2 TF-IDF Feature Extraction 6

5.3 Multinomial Naïve Bayes Classification 6

6. Justification of the Selected AI Techniques 6

7. System Design 7

8. Implementation Details 8

8.1 Dataset 8

8.2 Technologies Used 8

8.3 User Interface 8

9. Testing and Results 9

10. Limitations and Discussion 14

11. Ethical and Social Considerations 14

12. Conclusion 15

13. Individual Contribution 15

14. References 16

ABSTRACT

Phishing emails represent one of the major cyber security risks that try to lure users into sharing confidential information, clicking harmful links or doing anything else. Manual recognition of phishing emails proves to be a challenging task, especially for those who have little knowledge of cyber security. SafeMail AI is an AI-powered system that helps detect phishing emails.

The suggested system makes use of three Artificial Intelligence methodologies: Rule-Based Reasoning, TF-IDF feature extraction and classification using Multinomial Naïve Bayes. The rule-based part of the system looks for signs that are indicative of phishing, such as suspicious words, threatening tone, request for confidential information, suspicious URLs etc. The TF-IDF algorithm transforms the text content of the emails into numeric feature vectors and then the vectors are classified using the Multinomial Naïve Bayes classifier.

This ML model was created using a labelled data set that consisted of 82,486 emails, where 42,891 emails were identified as phishing emails, and 39,595 emails as legitimate emails. This dataset was split as 80% for training and 20% for testing using stratified sampling. A web-based application using Streamlit library has been created which lets the user input their email and get the classification output with the confidence score.

The developed prototype shows that the use of rule-based reasoning along with machine learning algorithms can create an intelligent system for detecting phishing emails.

1. Introduction

Email is a popular medium for personal, education, and business communication. But email is also extensively utilized by the attackers for carrying out phishing attacks. The emails that are used for executing phishing attacks are actually fake emails that trick people into doing some unsafe activities.

Detecting phishing emails manually might be quite hard because the scammers usually try to make phishing emails look authentic. This might be done by mimicking some real organizations, creating

urgency, asking for personal data, or including malicious links. People who do not have enough knowledge about cybersecurity might find it rather hard to recognize such emails.

Artificial Intelligence could aid in the detection of phishing attacks by performing automatic analysis of e-mails and discovering patterns associated with the e-mail message. The machine learning approach can use existing labeled datasets of phishing and legitimate e-mails to discover patterns, whereas rule-based reasoning uses predefined rules.

SafeMail AI has been designed as an advanced AI-based hybrid approach for phishing detection in emails, using Rule-Based Reasoning, TF-IDF Feature Extraction, and Multinomial Naïve Bayes Classification. SafeMail AI analyses the email text and gives phishing or non-phishing classification along with explanations for the same.

2. Problem Background

The issue of phishing is still topical for cybersecurity as the emails themselves can be used to get access to people's passwords, information related to banks, credit cards, one-time passwords, and any other sensitive information. In addition, phishing attack can also persuade users to click malicious links and do dangerous actions.

Traditionally, rule-based systems of phishing detection work based on pre-established rules for detection of known indicators of phishing emails. For example, an email with suspicious phrases, urgent appeals, requests for some sensitive information or with suspicious links can be considered as a potentially malicious one. The drawback of such approach consists in the fact that the changing of email content by attackers can make such rules ineffective.

The machine learning approach allows learning the patterns by studying the examples of phishing and legitimate emails. Text processing can be performed by means of NLP techniques, while the transformation of textual data into numbers suitable for further classification can be performed with the help of feature-extraction methods such as TF-IDF.

For these reasons, SafeMail AI applies both rule-based reasoning and machine-learning classification.

3. Problem Statement

The use of phishing emails is a cybersecurity threat since they seek to make the user reveal confidential data, visit harmful links, or conduct unsafe actions. Manual phishing detection relies on the cybersecurity awareness of the user and, thus, may not always be reliable.

The rule-based methods allow for detecting phishing based on known criteria like suspicious keywords, threats and warnings, requests for sensitive data, and suspicious links. However, these fixed rules cannot help detect the phishing emails if the attacker changes the phrasing or structuring of an email.

SafeMail AI resolves this issue by employing Rule-Based Reasoning, TF-IDF Feature Extraction, and Multinomial Naïve Bayes Classification. The system examines the content of an email and classifies it as either Phishing or Legitimate with the supporting details like the probability of being phishing and suspicious features.

4. Objectives

4.1 Main Objective

The primary aim of SafeMail AI is to create a highly intelligent system based on AI which will analyze the contents of the email and help in detecting any phishing emails.

4.2 Specific Objectives

1. Collection and preprocessing of a labeled data set comprising of phishing and legitimate emails.
2. Feature extraction of key textual features from the email content based on TF-IDF approach.
3. Model development of Multinomial Naïve Bayes classification to classify the emails into phishing and legitimate categories.
4. Implementation of IF-THEN rules for detection of known indicators of phishing, such as suspicious words, threats, sensitive information, and suspicious URL.
5. Web application development using Streamlit for the implementation of the proposed components of AI.
6. Evaluation of the proposed system through sample email data sets.

5. Selected AI Techniques

SafeMail AI employs three main AI techniques:

5.1 Rule-Based Reasoning

Rule-Based Reasoning is employed to find defined phishing features with the help of IF-THEN rules. The system evaluates the submitted email based on four general types of features:

- Suspicious keywords
- Urgent or threatening phrases
- Requests for confidential data
- Suspicious URLs

Each found feature adds one point to the rule score. If there are two or more features, the rule-based module marks the email as suspicious.

5.2 TF-IDF Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF) transforms the preprocessed email text into numerical feature vectors, which enables processing textual data by the machine learning classifier.

5.3 Multinomial Naïve Bayes Classification

The Multinomial Naïve Bayes Classifier is trained based on the labelled phishing and legitimate emails. The TF-IDF vectors are provided to the classifier to make it learn text features of both emails.

6. Justification of the Selected AI Techniques

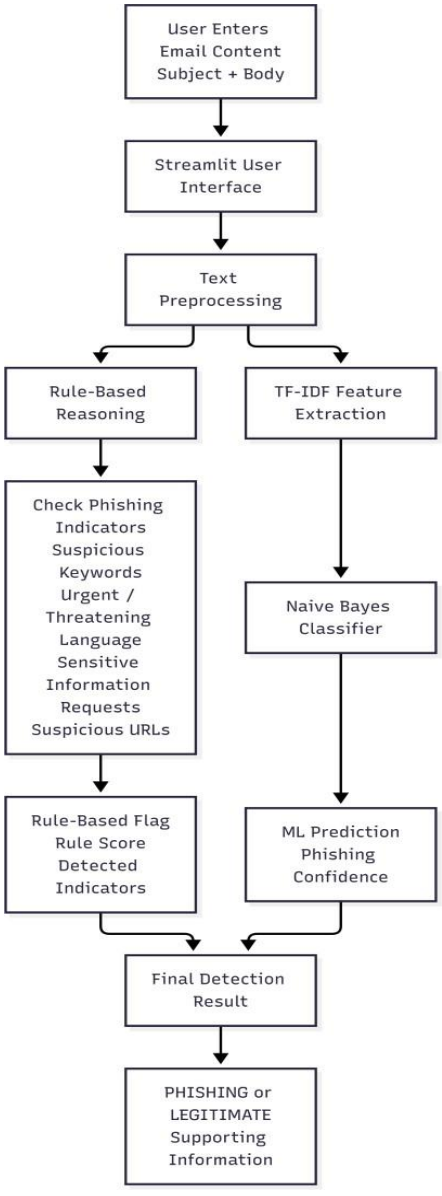
Rule-Based Reasoning has been chosen due to the fact that many common features of phishing can be easily presented through IF-THEN rules. It will make the system more understandable for the user as he will know the features detected by the system.

TF-IDF has been chosen as machine learning algorithms cannot deal with the textual content of emails directly. TF-IDF offers a way to present the importance of words and phrases numerically and hence is a good choice for classification of text.

Multinomial Naïve Bayes algorithm has been chosen due to its efficiency and suitability for text classification. This algorithm can be trained using TF-IDF features and is good for the implementation within the given ten weeks development time frame.

7. System Design

Original e-mail is processed using rule-based technique and the sanitized e-mail using TF-IDF and Multinomial Naïve Bayes techniques. Classification is Phishing if the model predicts phishing or the rule score is at least two. Otherwise, it is classified as Legitimate.



8. Implementation Details

8.1 Dataset

SafeMail AI uses a labelled dataset containing 82,486 email records:

Email Class	Label	Number of Emails
Phishing	1	42,891
Legitimate	0	39,595
Total		**82,486**

The dataset comprises a fairly balanced ratio of phishing and non-phishing emails. Data cleaning is done before model training using the email dataset. The dataset is split in a ratio of 80% training set and 20% test set by utilizing stratified sampling.

8.2 Technologies Used

The project uses Python as the main programming language. Pandas is used for loading and preprocessing the data set. Scikit-learn is used for TF-IDF feature extraction, Multinomial Naïve Bayes classification, and evaluation of the model.

The rule-based system is implemented using Python, and Streamlit is used for creating the web interface. Google Colab or Jupyter Notebook can be used in the process of creating the model and experimenting. Matplotlib is used to display the evaluation results like the confusion matrix. GitHub is used for version control and team cooperation.

8.3 User Interface

The interface of Streamlit enables a user to input email content for analysis. The output displayed after analyzing the inputted email is:

Classification result: Phishing/Legitimate

Verdict by ML: Classification by model

Confidence of ML: Phishing confidence value

Rule flagging: If rules identify the email as suspicious

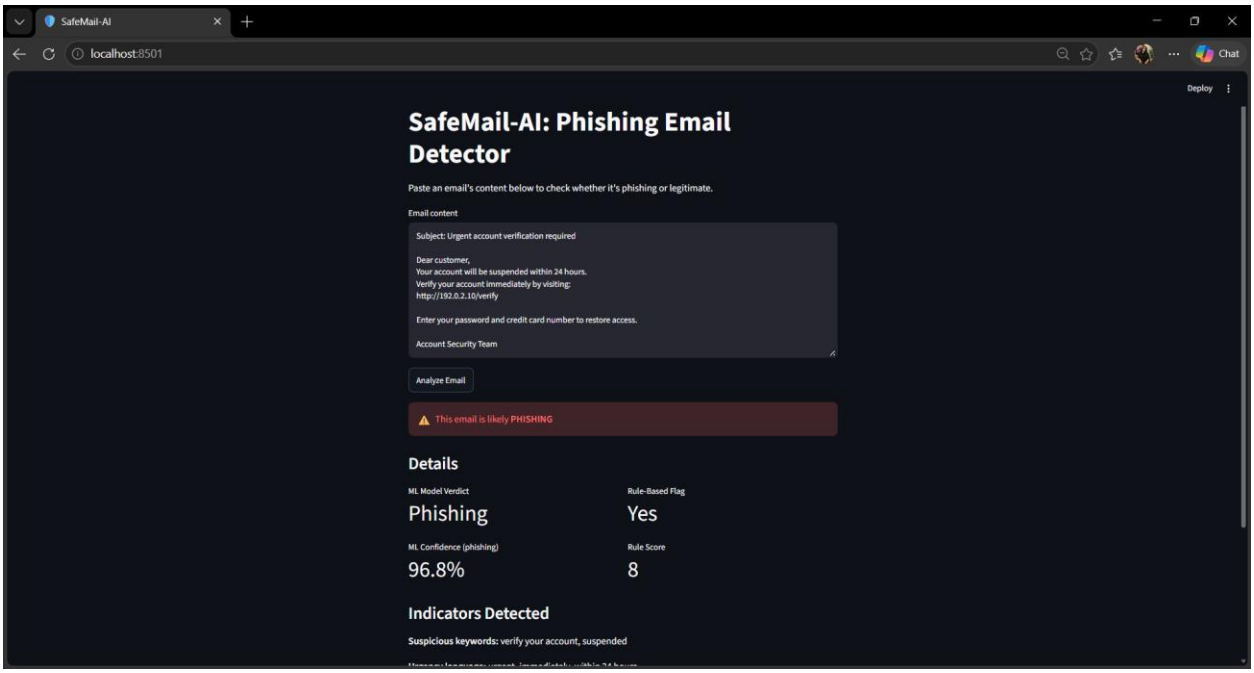
Score by rule: Number of indicators detected

Indicators detected: Description of suspicious indicators detected

9. Testing and Results

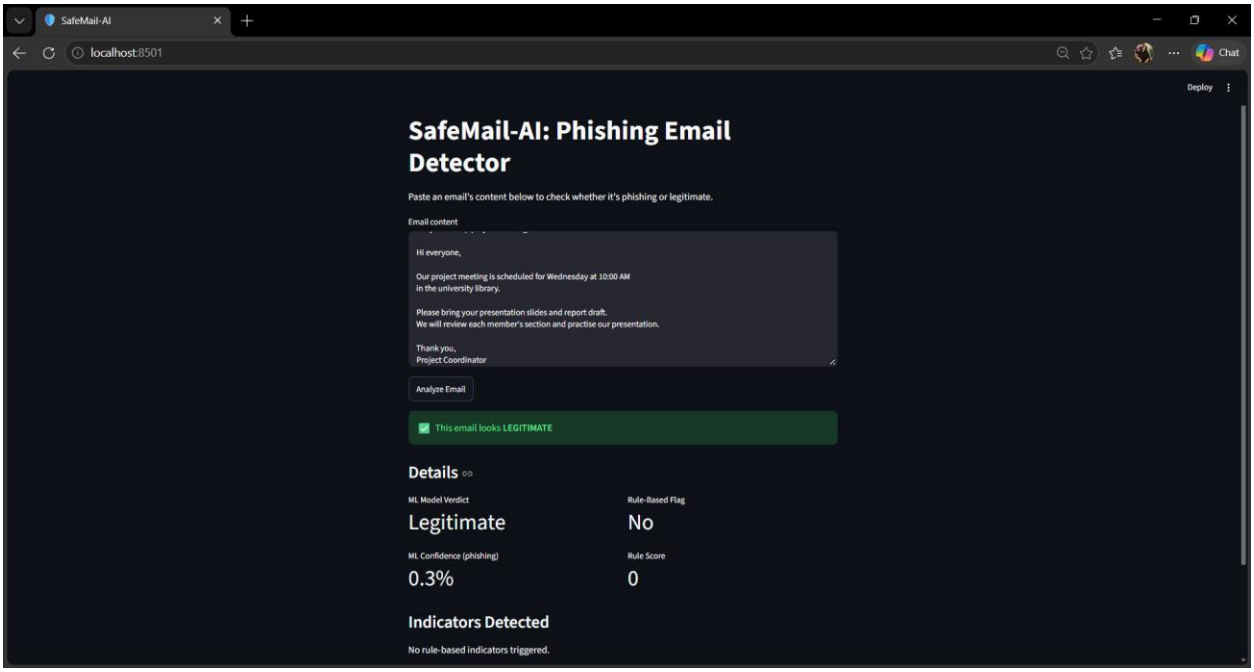
Naïve Bayes classification was assessed on the test set comprising 20% of the data. The entire SafeMail AI system was tested by using emails classified as phishing or legitimate and null values.

- Phishing email test input and detection result



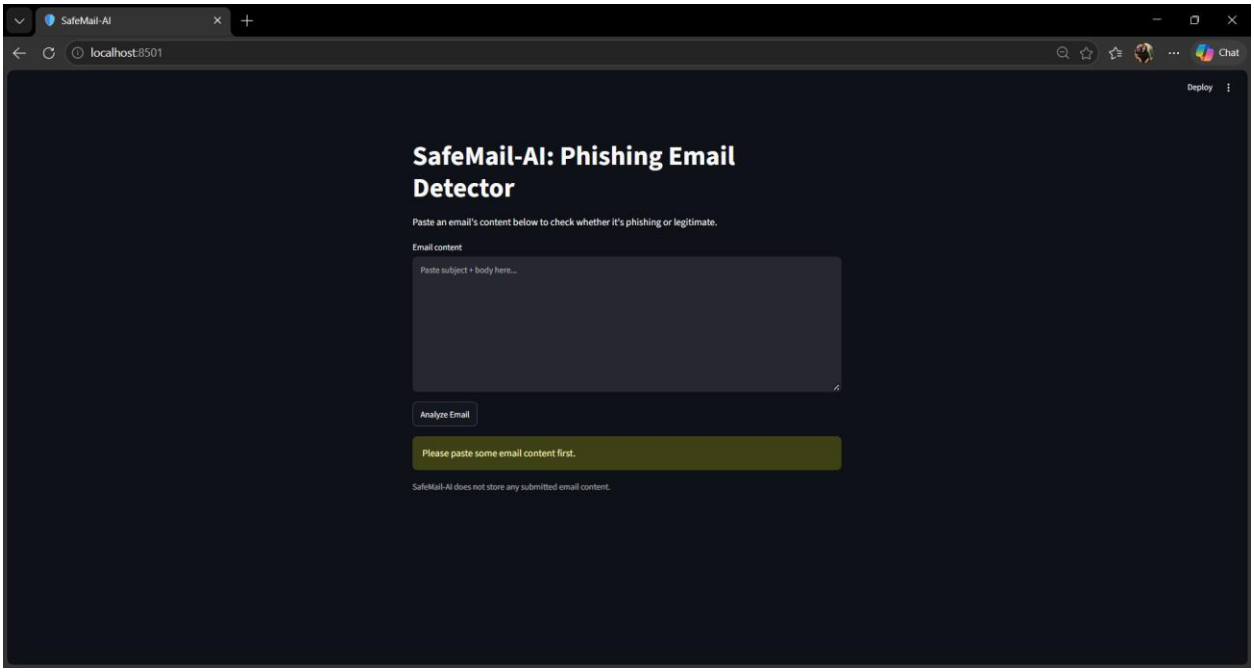
The system correctly classified the email as Phishing, with an estimated ML phishing probability of 96.8% and a rule score of 8. Both the model and rule-based component identified the email as suspicious.

- Legitimate email test input and detection result



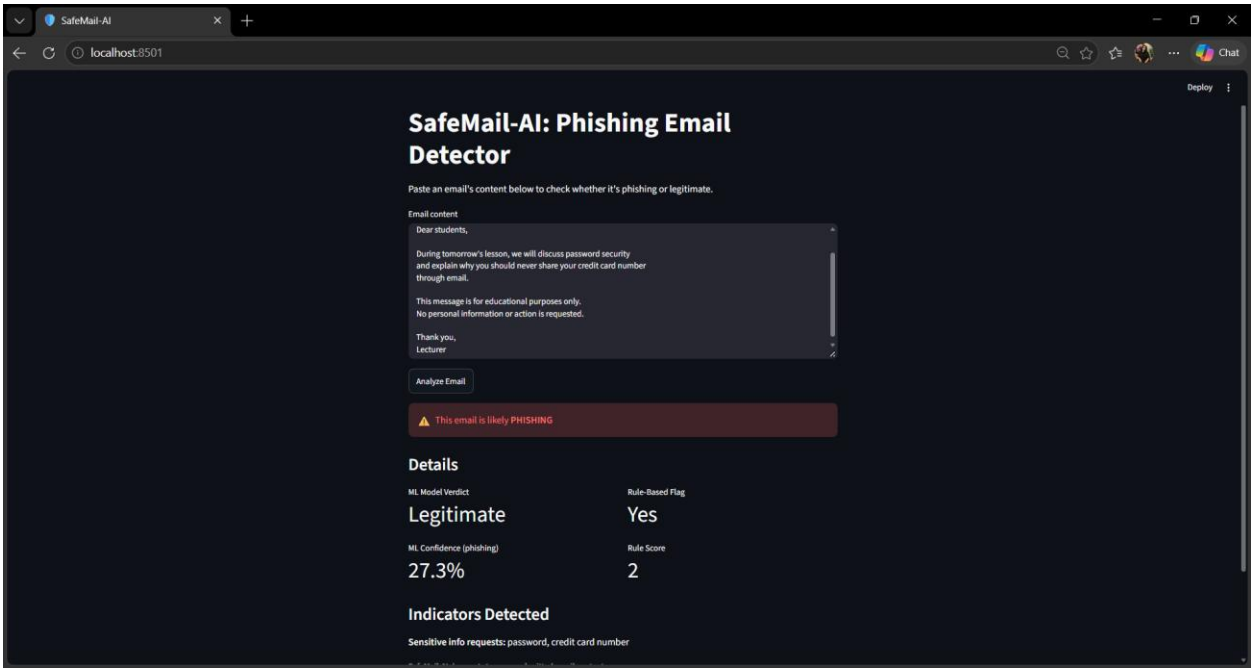
The system correctly classified the group-project meeting announcement as Legitimate, matching the expected result.

- Validation warning for empty email input



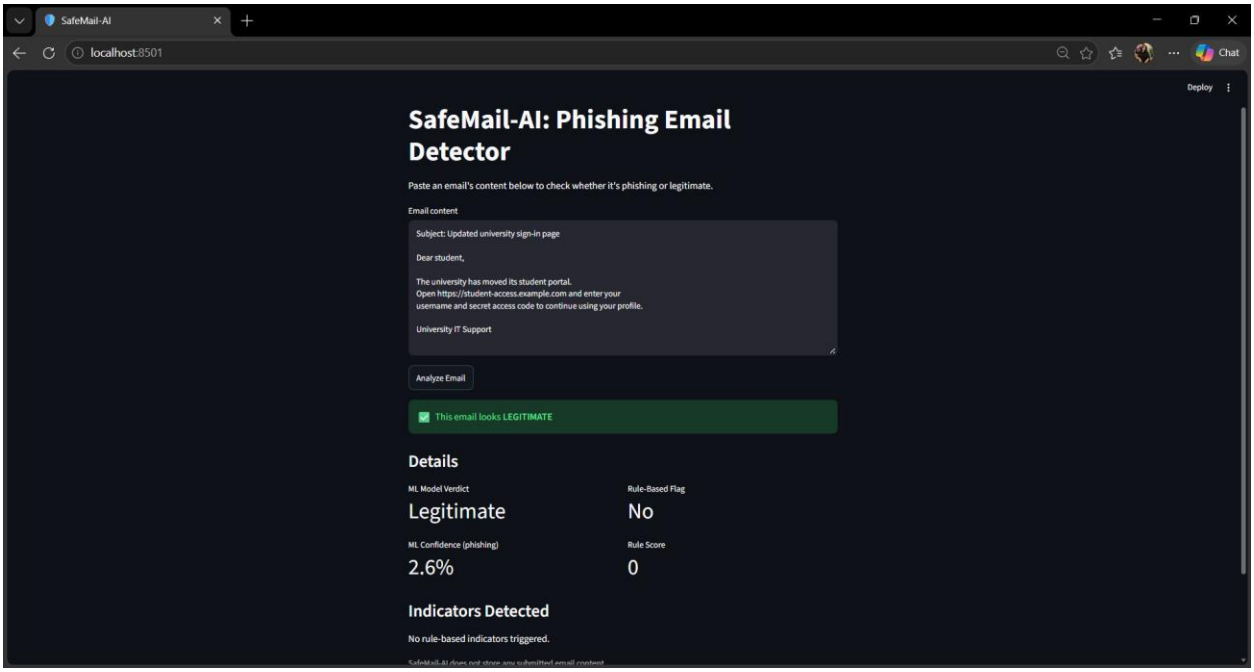
When no email content was entered, the system displayed “Please paste some email content first.” This confirms that empty input is handled before classification.

- Legitimate educational email incorrectly flagged by keyword rules



The ML model predicted Legitimate, with an estimated phishing probability of 27.3%. However, the rules detected “password” and “credit card number,” producing a score of 2 and overriding the model’s verdict. The final Phishing classification was a false positive, demonstrating that keyword rules do not understand educational context.

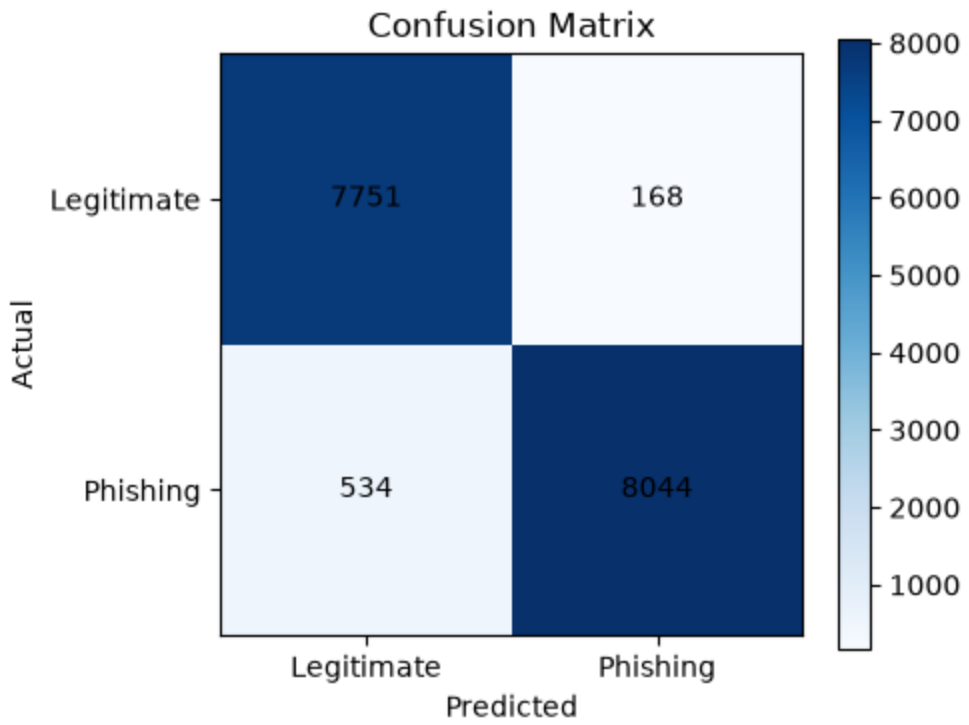
- Phishing email test with fewer predefined indicators



The fabricated phishing email was, however, wrongly detected as Legitimate, where its ML phishing probability is estimated at 2.6% while the rule score is 0. The erroneous detection shows how the algorithm can miss deceptive emails that do not contain any predetermined phrases.

- Naïve Bayes Model Evaluation

Metric	Result
Accuracy	95.74%
Phishing precision	97.95%
Phishing recall	93.77%
Phishing F1 score	95.82%



The Naïve Bayes classification tool identified 7,751 authentic emails and 8,044 phishing emails. The classifier generated 168 false positives and 534 false negatives. These results only represent performance of the classifier itself. Application tests have also indicated that the entire system can generate a false positive due to the keyword rules and miss a phishing attempt.

10. Limitations and Discussion

- Contextual limitations: The guidelines rejected a legitimate educational message since the terms “password” and “credit card number” were included, despite the prediction of Legitimate by the ML model.
- Missed phishing: The fictitious portal redirection message was recognized as Legitimate, which is an example of how misleading messages can bypass learned and predefined patterns.
- Just URL matching: The system analyzes URL patterns but does not verify website reputation and examines webpages.
- Dependency on data set: The performance of the model is limited by the quality of the data set used for training.
- Scope of evaluation: The accuracy, precision, recall, and F1 score only evaluate the Naïve Bayes classifier performance and not the hybrid system as a whole.

11. Ethical and Social Considerations

Data Privacy

Emails can be personal and sensitive in nature. Hence, it is not advisable to store unnecessary information. The system must analyze only the data needed for detecting phishing attacks.

Accuracy and Reliability

It cannot be claimed that an AI classifier will always classify emails accurately. Hence, the user must use the SafeMail AI as an assistant in the decision-making process.

Bias and Dataset Limitations

Since the machine learning model depends on the training dataset, if there is any bias in the training dataset, the model will be affected by that.

Human Responsibility

The tool is built to support the users; however, humans need to take care of themselves as well. Hence, it is recommended to be careful while using emails marked as legitimate by the tool.

Misuse and Automated Decisions

The output from the classifier is never considered as evidence for labeling the emails as spam or phishing ones.

12. Conclusion

SafeMail AI uses Rule-Based Reasoning, TF-IDF Feature Extraction, and Multinomial Naïve Bayes Classification to detect phishing emails using the Streamlit app. The output of the model is the final classification, machine learning phishing probability, score of the rules applied, and indicators of phishing found. The accuracy of the Naïve Bayes classifier on the holdout test data is 95.74%. However, application testing showed both a false positive due to keyword rules and a missing phishing case. Thus, SafeMail AI should assist users to be aware rather than provide safety.

13. Individual Contribution

Group Member	Role	Contribution
TPSD Perera (D/BIT/24/0089)	Backend Development and System Integration	Developed core system components and supported integration of the rule-based component, TF-IDF feature extraction, Naïve Bayes model, and Streamlit interface.
AHS Venushika (D/BCS/25/0024)	Dataset and Preprocessing	Contributed to dataset preparation and preprocessing and prepared phishing and legitimate email samples for feature extraction and model training.
DPR Pabasara (D/BIT/24/0007)	Project Coordination and Documentation	Coordinated project progress, maintained project documentation and reports, and performed system testing and verification of phishing and legitimate detection results.
SAED Eleesha (D/BSE/25/0015)	AI Model Development	Contributed to activities related to the Naïve Bayes classifier and rule-based phishing detection and assisted with model-related project work.

14. References

[1] S. A. Salloum, T. Gaber, S. Vadera, and K. Shaalan, “Phishing email detection using natural language processing techniques: A literature survey,” Procedia Computer Science, vol. 189, pp. 19–28, 2021. Available: https://doi.org/10.1016/j.procs.2021.05.077

[2] S. A. Salloum, T. Gaber, S. Vadera, and K. Shaalan, “A systematic literature review on phishing email detection using natural language processing techniques,” IEEE Access, vol. 10, pp. 65703–65727, 2022. Available: https://doi.org/10.1109/ACCESS.2022.3183083

[3] A. Alhogail and A. Alsabih, “Applying machine learning and natural language processing to detect phishing email,” Computers & Security, vol. 110, Art. no. 102414, Nov. 2021. Available: https://doi.org/10.1016/j.cose.2021.102414

[4] G. Sonowal, “Phishing email detection based on binary search feature selection,” SN Computer Science, vol. 1, no. 4, Art. no. 191, 2020. Available: https://doi.org/10.1007/s42979-020-00194-z

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[6] I. Fette, N. Sadeh, and A. Tomasic, “Learning to detect phishing emails,” in Proceedings of the 16th International Conference on World Wide Web (WWW ’07), pp. 649–656, 2007. Available: https://doi.org/10.1145/1242572.1242660

ESSENTIALS OF ARTIFICIAL INTELLIGENCE
INDIVIDUAL CONTRIBUTION SHEET

Student Name	DPR Pabasara	**Registration No.**	D/BIT/24/0007
Degree / Intake	IT-41	**Group No.**	49
Project Title	SafeMail AI – AI Based Phishing Email Detector		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Project documentation	Prepared all project documentation, including the proposal, Progress Review 1 report, and complete final report.	Microsoft Word chatGPT	Proposal PR1 report Final report
2. System testing and analysis	Performed all system testing, captured screenshots, and explained successful detections, false positives, false negatives, and empty-input validation.	Streamlit, manual functional testing	Test screenshots and Testing and Results section
3. README documentation	Added the README run instructions to GitHub through a documentation branch and merged pull request.	Git GitHub	README.md and merged pull request
4. Presentation preparation	Created the project presentation slides using Claude for assistance and PowerPoint for preparation and formatting.	Claude Microsoft PowerPoint	Final presentation slides
5. Recorded presentation	Presented and recorded my assigned section using PowerPoint's recording feature.	Microsoft PowerPoint Recorder	My recorded presentation clip

6. Final video editing	Edited and combined all four members' presentation recordings into one final video.	Adobe Premiere Pro	Complete final presentation MP4
7. Project coordination and team guidance	Guided the team throughout the project, coordinated tasks, followed up on progress, and helped members prepare and organize their work for the final submission.	Group communication and progress tracking	Task discussions, progress updates, and organized final submission material

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** __15/10/2026_____

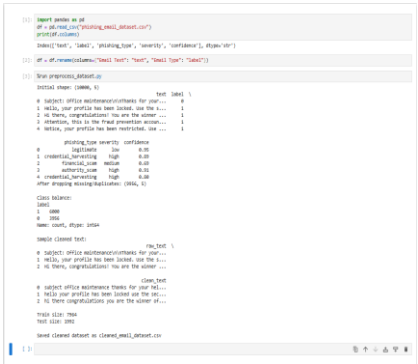
Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**AHS Venushika**	**Registration No.**	**D/BCS/25/0024**
Degree / Intake	**Computer Science**	**Group No.**	**49**
Project Title	**AI-Based Phishing Email Detection System Using Rule-Based Reasoning and Naïve Bayes Classification**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Dataset collection	Collected labelled phishing and legitimate email samples from a public dataset to use for training and testing.	Kaggle phishing email dataset, Python, Pandas	phishing_email_dataset.csv
Data cleaning and preprocessing	Removed missing and duplicate records, checked class balance, and cleaned the email text (lowercasing, removing HTML tags, URLs and punctuation).	Python, Pandas, regular expressions (text preprocessing)	 preprocess_dataset.py
Dataset preparation for feature extraction and model training	Prepared the cleaned text and labels for TF-IDF feature extraction and split the data into training and testing sets for the Naïve Bayes model.	Scikit-learn (train/test split), Jupyter Notebook	cleaned_email_dataset.csv

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**SAED Perera**	**Registration No.**	**D/BSE/25/0015**
Degree / Intake	**Intake 42 -BSc (Hons) in Software Engineering**	**Group No.**	**Group 49**
Project Title	**SafeMail-AI :Phishing Email Detection System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Machine Learning Model Training & Pipeline	Developed, trained, and evaluated the core phishing detection model using Multinomial Naïve Bayes and TF-IDF feature extraction. Handled train-test data splitting (80/20) and saved serialized model assets (.pkl).	Python, Scikit-learn, Pandas, Pickle, TF-IDF Vectorizer, Multinomial Naïve Bayes (MultinomialNB)	Source Code (04_train_model.py), Model Artifacts (nb_model.pkl, tfidf_vectorizer.pkl)
Hybrid Rule-Based Detection Engine	Designed and implemented a rule-based risk evaluation module to flag suspicious indicators (urgency language, sensitive info requests, raw IP URLs, and link shorteners).	Python, Regular Expressions (re), Custom Logic Architecture	Source Code (rule_based.py, test_rules.py)
Model Evaluation & Testing	Built manual and automated evaluation scripts to validate model performance using accuracy, precision, recall, F1-score, and confusion matrix visualization.	Matplotlib, Scikit-learn Metrics	Confusion Matrix (confusion_matrix.png), Evaluation Script (05_manual_eval.py)
Version Control & GitHub Collaboration	Created feature branches (eleesha-model), committed model scripts/assets, and	Git, GitHub	GitHub Repository Commits & Pull Requests (safemail-ai repo)

	managed Pull Requests for team integration.		

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: Eleesha **Date:** 2026/10/17

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	**TPSD Perera**	**Registration No.**	**D/BIT/24/0089**
Degree / Intake	**Intake 42 -BSc (Hons) in Information Technology**	**Group No.**	**Group 49**
Project Title	**SafeMail-AI :Phishing Email Detection System**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Backend Prediction Pipeline	Designed and implemented the core predict_email() function. It takes raw email text, runs it through the rule-based engine and the trained Naïve Bayes model, and returns one structured result (final label, ML prediction, confidence score, triggered indicators).	Python, Pickle (loading saved model assets), Scikit-learn (TF-IDF vectorizer and Naïve Bayes inference)	Source Code (predictor.py)
Hybrid Decision Logic	Built the logic that combines both detection methods into one verdict. An email is flagged as phishing if either the ML model or the rule-based system detects it. This lets each method catch what the other misses.	Python, Hybrid Rule-Based + Machine Learning Architecture	Source Code (predictor.py), System architecture diagram
Version Control & Repository Management	Created and set up the GitHub repository (safemail-ai), configured .gitignore, added collaborators, and created the perera-integration branch. Managed Pull Requests and merged all four team branches into main in dependency order.	Git, GitHub	GitHub Repository Commits, Branches & Pull Requests (safemail-ai repo)

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** 2026/10/19

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.